Using the Risk Analysis Spreadsheet Tool

This section provides step-by-step instructions for using the spreadsheet-based **Risk Analysis Tool** that accompanies this report. It outlines the tool's data requirements, explains how to operate the tool, and describes how to interpret the results of the analysis. A worked example is presented at the end of the section to demonstrate the tool's application in practice. The Risk Analysis Tool is designed to support quantitative risk analysis within a Transportation Asset Management (TAM) framework. It enables the assessment of various transportation assets in relation to different natural hazards considered as risk factors. The tool estimates the risk priority for each asset, along with the return on investment (ROI) and its payback period, by integrating hazard probability, potential consequence costs, and the vulnerability of each asset. The results are summarized using four key metrics: the total risk value, risk priority, ROI, and payback period. The tool has been tested using data from the state of Utah, which is presented in the case study included in this report.

System Requirements

The spreadsheet-based Risk Analysis Tool is an Excel workbook designed to run in Microsoft Excel. The tool has been tested on both Windows and macOS operating systems.

Tool Components

The Risk Analysis Tool consists of five worksheets: one input worksheet (**Inputs and Additional Parameters**), two calculation worksheets (**Vulnerability and Risk Analysis**), one result worksheet (**Summary Results**), and one manual worksheet (**Instruction**). Certain fields in the Risk Analysis Tool are hidden to streamline its use; however, there are no locked fields or protected calculations, and the tool does not utilize any plug-ins or macros.

Using the Risk Analysis Tool

Users of the Risk Analysis Tool enter data in the **Inputs and Additional Parameters** worksheet, select the desired assets in the **Risk Analysis** worksheet, and then navigate between worksheet tabs to view calculation details and analysis results. The model includes several fixed default parameters, which users can override at their discretion.

• *Inputs and Additional Parameters Worksheet*

The **Inputs and Additional Parameters** worksheet contains the necessary parameters for performing a risk analysis. It includes general information about the asset inventory—such as asset types, geographical location, operational parameters, identified threats, and their probabilities—as well as additional parameters developed in this study: asset criticality, exposure and sensitivity to each threat, and the asset's adaptive capacity in the event of damage. A partial view of the Inputs and Additional Parameters worksheet is shown in **Figure 1**.

Target ID,FULLNAME	Target ID	FULLNAME	Asset Type	Geographical Location	Repair Cost	Replacement Cost	AADT	TADT	Delay Time	Adaptive Capacity	Criticality_RSL	RF_Prob	RF_Exposure	RF_Sensitivity	RF_RequiredRepairDay
122254,HWY 6	122254	HWY 6	Approach	(-110.8598,39.638814)	225720	1128600	13000	1963	1440	Low	0.00009288	0	Not Close	Low	0
122255,HWY 6	122255	HWY 6	Approach	(-110.843889,39.616094)	221284.8	1106424	13000	1963	1	High	0.0000254	0	Not Close	Low	0
122256,HWY 6	122256	HWY 6	Approach	(-110.852783,39.617187)	560181.6	2800908	13000	1963	0	High	0.00002535	0	Not Close	Low	0
122257,HWY 6	122257	HWY 6	Approach	(-110.343114,39.264459)	95040	475200	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122258,HWY 6	122258	HWY 6	Approach	(-110.338824,39.240172)	560181.6	2800908	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122259,HWY 6	122259	HWY 6	Approach	(-110.337418,39.135919)	202215.288	1011076.44	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122260,HWY 6	122260	HWY 6	Approach	(-110.868273,39.710562)	90024	450120	9800	4067	230	High	0.00004443	0	Not Close	Low	0
122261,HWY 6	122261	HWY 6	Approach	(-112.499104,39.400436)	105424	527120	2600	772	1041	Low	0.00005602	0	Not Close	Low	0
122262,HWY 6	122262	HWY 6	Approach	(-112.500715,39.399105)	560181.6	2800908	2600	772	249	High	0.00001888	0	Not Close	Low	0
122263,MAIN ST	122263	MAIN ST	Approach	(-110.687017,39.542576)	560181.6	2800908	7500	3038	475	High	0.00004761	0	Not Close	Low	0
122264,HWY 6	122264	HWY 6	Approach	(-110.766814,39.560522)	560181.6	2800908	11000	3388	9	High	0.00003157	0	Not Close	Low	0
122265,HWY 6	122265	HWY 6	Approach	(-111.200026,39.933228)	349800	1749000	9200	3873	1440	Low	0.00009942	0	Not Close	Low	0
122266,HWY 6	122266	HWY 6	Approach	(-112.625428,39.330513)	560181.6	2800908	3500	924	4	High	0.00000923	0	Not Close	Low	0
122267,MAIN ST	122267	MAIN ST	Approach	(-110.700054,39.542557)	560181.6	2800908	9100	3276	7	High	0.00002876	0	Not Close	Low	0
122268,HWY 6	122268	HWY 6	Approach	(-110.298363,39.024573)	560181.6	2800908	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122269,HWY 6	122269	HWY 6	Approach	(-111.589202,40.082508)	258808	1294040	12000	3252	5	High	0.00003172	0	Not Close	Low	0
122270,HWY 6	122270	HWY 6	Approach	(-111.24138,39.956061)	560181.6	2800908	9200	3873	174	High	0.00004005	0	Not Close	Low	0
122271,HWY 6	122271	HWY 6	Approach	(-111.502389,40.027875)	560181.6	2800908	12000	3252	204	High	0.00004106	0	Not Close	Low	0
122272,HWY 6	122272	HWY 6	Approach	(-111.216788,39.949213)	276575.2	1382876	9200	3873	199	High	0.00004123	0	Not Close	Low	0
122273,HWY 6	122273	HWY 6	Approach	(-111.241768,39.956244)	186032	930160	9200	3873	174	High	0.00004005	0	Not Close	Low	0
122274,HWY 6	122274	HWY 6	Approach	(-110.860298,39.652552)	560181.6	2800908	14000	3598	5	High	0.00003587	0	Not Close	Low	0
122275,HWY 6	122275	HWY 6	Approach	(-111.166454,39.931401)	137280.264	686401.32	9200	3873	1440	Low	0.00009942	0	Not Close	Low	0
122276,HWY 6	122276	HWY 6	Approach	(-110.965916,39.831597)	390852	1954260	7700	3211	216	High	0.00003665	0	Not Close	Low	0
122277,HWY 6	122277	HWY 6	Approach	(-110.857892,39.691745)	746592	3732960	9300	3367	0	High	0.00002916	0	Not Close	Low	0
122278,HWY 6	122278	HWY 6	Approach	(-110.996099,39.843708)	101200	506000	7700	3211	339	High	0.00004242	0	Not Close	Low	0
122279,HWY 6	122279	HWY 6	Approach	(-111.635682,40.115234)	572176	2860880	32000	4576	1	High	0.00006101	0	Not Close	Low	0
122280,HWY 6	122280	HWY 6	Approach	(-111.242227,39.956428)	186040.8	930204	9200	3873	174	High	0.00004005	0	Not Close	Low	0
122281,SHEEP CREEK RD	122281	HEEP CREEK R	Approach	(-111.340528,39.974665)	560181.6	2800908	0	0	0	High	0	0	Not Close	Low	0
122282,HWY 6	122282	HWY 6	Approach	(-110.938791,39.815773)	92285.248	461426.24	7700	3211	222	High	0.00003693	1	Close	Low	0
122283,HWY 6	122283	HWY 6	Approach	(-111.803815,39.980578)	251504	1257520	4800	1224	2	High	0.00001226	0	Not Close	Low	0
122284,HWY 6	122284	HWY 6	Approach	(-110.860962,39.637075)	560181.6	2800908	13000	1963	0	High	0.00002535	0	Not Close	Low	0
122285,HWY 6	122285	HWY 6	Approach	(-110.823113,39.595586)	161568	807840	12000	2436	1440	Low	0.00009443	0	Not Close	Low	0
122286,HWY 6	122286	HWY 6	Approach	(-110.859889,39.639395)	225719.648	1128598.24	13000	1963	0	High	0.00002535	0	Not Close	Low	0
122287,HWY 6	122287	HWY 6	Approach	(-111.037149,39.876153)	560181.6	2800908	7700	3211	270	High	0.00003918	0	Not Close	Low	0
122288,HWY 6	122288	HWY 6	Approach	(-111.580488,40.07529)	560181.6	2800908	12000	3252	203	High	0.00004101	0	Not Close	Low	0
122289,HWY 6	122289	HWY 6	Approach	(-111.021112,39.857543)	560181.6	2800908	7700	3211	339	High	0.00004242	0	Not Close	Low	0
122290,HWY 6	122290	HWY 6	Approach	(-110.997735,39.844097)	206096	1030480	7700	3211	339	High	0.00004242	0	Not Close	Low	0
122291,HWY 6	122291	HWY 6	Approach	(-110.789248,39.587105)	405521.776	2027608.88	11000	3388	29	High	0.00003251	0	Not Close	Low	0
122292,HWY 6	122292	HWY 6	Approach	(-110.344519,39.26808)	560181.6	2800908	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122293,MAIN ST	122293	MAIN ST	Approach	(-112.58687,39.352459)	173008	865040	6100	1659	1	High	0.00001609	0	Not Close	Low	0
122294,HWY 6	122294	HWY 6	Approach	(-110.368784,39.331126)	560181.6	2800908	5800	2581	1440	Low	0.00008842	0	Not Close	Low	0
122295,HWY 6	122295	HWY 6	Approach	(-110.792735,39.587322)	480269.152	2401345.76	13000	3328	0	High	0.00003302	0	Not Close	Low	0
122296,HWY 6	122296	HWY 6	Approach	(-110.494865,39.46746)	560181.6	2800908	6300	2829	214	High	0.00003287	0	Not Close	Low	0
122297,HWY 6	122297	HWY 6	Approach	(-110.479985,39.454658)	560181.6	2800908	6300	2829	214	High	0.00003287	0	Not Close	Low	0
122298,HWY 6	122298	HWY 6	Approach	(-110.867498,39.697716)	560181.6	2800908	9800	4067	0	High	0.00003365	0	Not Close	Low	0
122299,HWY 6	122299	HWY 6	Approach	(-110.827336,39.600973)	274717.696	1373588.48	12000	2436	0	High	0.00002691	0	Not Close	Low	0
122300,HWY 6	122300	HWY 6	Approach	(-110.844309,39.615859)	221284.8	1106424	13000	1963	0	High	0.00002535	0	Not Close	Low	0
122301,HWY 6	122301	HWY 6	Approach	(-110.233219,38.98172)	427328	2136640	5800	2581	0	High	0.00002089	0	Not Close	Low	0
122302,MAIN ST	122302	MAIN ST	Approach	(-110.730929,39.542434)	560181.6	2800908	9100	3276	9	High	0.00002885	0	Not Close	Low	0
122303,HWY 6	122303	HWY 6	Approach	(-112.621712,39.33049)	206096	1030480	3500	924	4	High	0.00000923	0	Not Close	Low	0
122304,HWY 6	122304	HWY 6	Approach	(-110.575305,39.526628)	202408.536	1012042.68	6300	2829	466	High	0.00004469	0	Not Close	Low	0

Fig 1. Partial View of the Inputs and Additional Parameters Worksheet

• *Vulnerability Calculation Worksheet*

In the transportation context, the vulnerability of transportation assets refers to the probability that a threat will result in the estimated consequences (Brashear and Jones, 2010). In other words, vulnerability is a function of an asset's or system's exposure to hazards, sensitivity to. those hazards, and adaptive capacity. *Exposure* refers to whether an asset or system is located in an area experiencing the direct effects of a threat; *sensitivity* describes how the asset or system performs when exposed; and *adaptive capacity* represents the ability to cope with existing variability or future impacts.

In this study, the capabilities of Bayesian Belief Networks (BBNs) are used to quantify the effects of potential hazards on the performance of transportation infrastructure and to predict the probability of damage to assets (Figure 2). BBNs help capture the uncertainties and interdependencies within a transportation network's response to extreme events. They are well-established tools for estimating the likelihood of a given scenario and its outcomes when contributing factors are appropriately defined and quantified.

A BBN typically consists of a set of nodes (representing random variables) and directed links that model the dependencies between these variables. Each node may function as a parent or a child, depending on the structure of the network. In this context, child nodes are influenced by parent nodes. As shown in Table 1, this study defines hazard, location, asset condition, and redundancy as parent nodes, which influence the three vulnerability components—exposure, sensitivity, and adaptive capacity—that together determine the damage level. Nodes that serve both as parents and children are referred to as intermediate nodes. The links between nodes form a directed acyclic graph, meaning that relationships flow in one direction only and no feedback loops or cycles are allowed within the network.

For a BBN with variables $(A_1, A_2, A_3, \ldots, A_n)$, the joint probability of occurrence of all variables is expressed as:

$$p(A1,A2,A3,...,An) = p(A1 \mid A2,A3,...,An)p(A2 \mid A3,...,An)p(A3 \mid A4,...,An)$$

By denoting a set of parent nodes, $PP(A_i)$, the joint probability equation can be written as:

$$p(A1,A2,A3,...,An) = \prod p(Ai \mid PP(Ai))$$

The random variable A_i can be represented in either a Boolean format (e.g., Yes/No, True/False) or a qualitative format. In this study, the intermediate nodes—**exposure**, **sensitivity**, and **adaptive capacity**—are described qualitatively (Table 2). For exposure, an asset is considered to be *close to the hazard source* if the probability of experiencing at least one hazard is greater than zero. Asset condition is assessed based on the number of days required for repair following a hazard event. If the required repair days is less than 3.5 days, the asset is considered to be in *good* condition. A repair duration between 3.5 to 30 days indicates a *fair* condition, while a duration exceeding 30 days reflects a *poor* condition. To quantify adaptive capacity, the delay time experienced by an asset during a hazard is used as an indicator. Assets with a delay time shorter than 480 minutes are considered to have high adaptive capacity.

To quantify the relationships between these variables and the probability of damage occurrence, **Conditional Probability Tables (CPTs)** are required for each node with parent nodes. These CPTs are populated using expert judgment, historical data, or a combination of both, allowing the Bayesian Belief Network (BBN) to perform probabilistic inference and support risk-informed

decision-making. Table 3 illustrates the potential scenarios and the corresponding probability of damage associated with each.

Fig 2. BBN Model Developed in the Current Study to Calculate the Vulnerability of Transportation Assets due to Natural Hazards

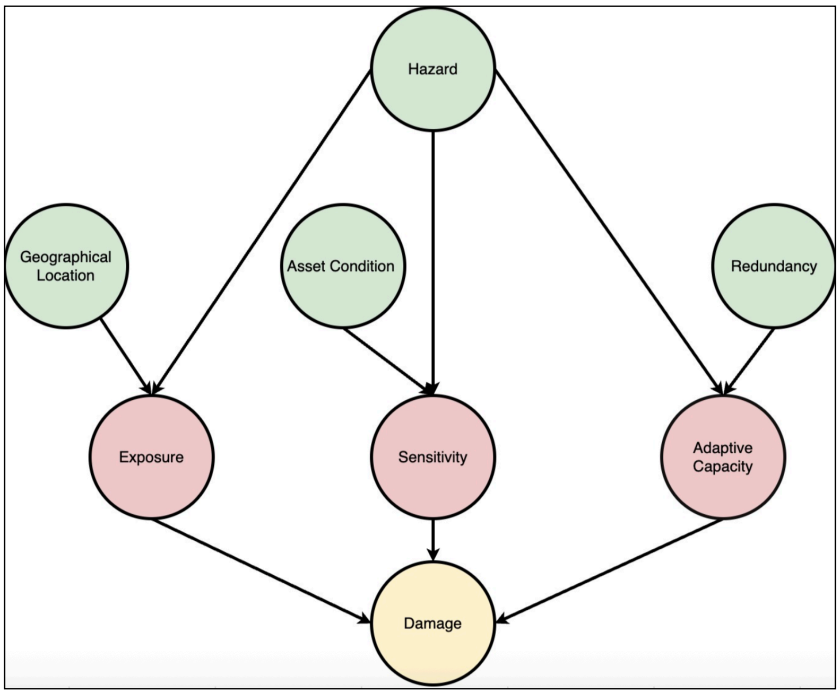

Table 1. Parent Nodes and the Potential Status

Parent Nodes	
Node	**State**
Hazard	Yes
	No
Geographical Location	Close
	Not Close
Asset Condition	Poor
	Fair
	Good
Redundancy	Low
	High

Table 2. Intermediate Nodes, the Potential Status, and the Probability of each State due to Hazard

Intermediate Nodes*		
Node	**Parent Nodes State**	**P(ci| H) (Statistical Data)****
Exposure	Close (Close to Hazard Source)	0.12
	Not Close (Far from Hazard Source)	0.88
Sensitivity	High (Poor Asset Condition)	0.05
	Moderate (Fair Asset Condition)	0.35
	Low,(Good Asset Condition)	0.6
Adaptive Capacity	Low (Redundancy Low)	0.16
	High (Redundancy High)	0.84

Table 3. Conditional Probability Table Developed for Vulnerability BBN Model

Conditional Probability Table (CPT)						
Scenario Description	**Exposure**	**Sensitivity**	**Adaptive Capacity**	**P(D|E,S,A)**	**P(E|H)×P(S|H)×P(A|H)**	**Vulnerability (P(D|H))*****
Close,High,High	Close	High	High	0.005	0.00096	**0.0000048**
Close,High,Low	Close	High	Low	0.001	0.00504	**0.00000504**
Close,Moderate,High	Close	Moderate	High	0.035	0.00672	**0.0002352**
Close,Moderate,Low	Close	Moderate	Low	0.006	0.03528	**0.00021168**
Close,Low,High	Close	Low	High	0.06	0.01152	**0.0006912**
Close,Low,Low	Close	Low	Low	0.011	0.06048	**0.00066528**
Not Close,High,High	Not Close	High	High	0.037	0.00704	**0.00026048**
Not Close,High,Low	Not Close	High	Low	0.007	0.03696	**0.00025872**
Not Close,Moderate,High	Not Close	Moderate	High	0.258	0.04928	**0.01271424**
Not Close,Moderate,Low	Not Close	Moderate	Low	0.05	0.25872	**0.012936**
Not Close,Low,High	Not Close	Low	High	0.442	0.08448	**0.03734016**
Not Close,Low,Low	Not Close	Low	Low	0.086	0.44352	**0.03814272**

• *Risk Analysis Worksheet*

The Risk Analysis worksheet contains a table that calculates the risk value and priority of the assets selected by users. Cells shaded in yellow are input cells. The outputs are calculated automatically in green cells as the inputs are updated. Blue cells include values extracted from other worksheets or tools developed as part of this study. Unshaded cells represent default values selected for this study, but they can be modified based on user requirements. The following procedure outlines how the tool calculates risk results for selected assets:

1. On the left side of the table, different asset categories are displayed. Users can select the *Structure ID* and *Full Name* of the desired asset from the dropdown lists (Figure 3).
2. Geographical location and certain asset attributes (AADT, TADT, replacement cost, and repair cost) are automatically populated from the *Inputs* worksheet. **Delay time** and **network criticality** are extracted from the GIS tool developed as part of this study (Figure 4).
3. Different hazard probabilities for the selected assets are automatically displayed from the *Inputs* worksheet (Figure 5).
4. Required repair days for each hazard is calculated based on Table 4.

Figure 3. Choosing Desired Asset

Asset Category	Structure ID & Full Name
Bridge Approach	

Figure 4. Geographical and Operational Parameters of the Asset

Geographical Location		Asset Attributes			
Longitude_X		AADT		Replacement Cost	
		TADT		Repair Cost	
Latitude_Y		Delay Time		Network Criticality *	

Figure 5. Natural Hazards and Their Occurrence Probability

Probability	
Hazard Type	Hazard Probability
Rockfall	
Flood	
Debris Flow	
Earthquake	
Avalanche	

Table 4. Required Days to Fix the Damage for Different Asset Types and Different Hazards

Asset	Earthquake	Debris Flow	Flood	Avalanche	Rock Fall
Bridge deck	30 days	3.5 days	3.5 days	0	0
Bridge approach	30 days	3.5 days	3.5 days	0	0
Box culvert	30 days	3.5 days	3.5 days	0	0
Pipe culvert	30 days	3.5 days	3.5 days	0	0
Pavement	30 days	3.5 days	3.5 days	0	0

5. Consequence includes direct and indirect consequences (Meyer et al., 2012; Eidsvig et al., 2021) (Figure 6). **Direct consequence** refers to losses resulting from physical damage to an asset due to a specific threat and is determined based on *Table 5*. **Indirect consequence** refers to losses triggered by the failure of transportation assets, reflecting the impact on network performance and human safety. It can be quantified by the monetary cost of additional travel time or distance caused by the asset's failure, using the following formula:

$$Indirect\ Consequence = [Delay\ Time \times Unit\ Value\ of\ Time\ per\ Passenger\ Car\ \times AADT + Unit\ Value\ of\ Time\ per\ Truck\ \times TADT\ ] \times Number\ of\ Days\ to\ Fix\ Damaged\ Asset + Impeded\ Flow\ Cost$$

Impeded flow occurs due to traffic control restrictions after the roadway is reopened to limited traffic flow following temporary repairs and it is calculated based on following formula:

$$Impeded\ flow\ cost\ = Asset\ Replacement\ Cost\ \times\ 0.33$$

Figure 6. Potential Consequence for Different Hazards

Consequence				
Required Repair Days	**Impeded Flow Cost**	**Direct Consequence**	**Indirect Consequence**	**Consequence**

Table 5. Direct Consequences for Various Types of Assets due to Different Hazards

Asset Type	Earthquake Direct Consequence	Debris Flow Direct Consequence	Flood Direct Consequence	Avalanche Direct Consequence	Rock Fall Direct Consequence
Bridge deck	Replacement cost	Replacement cost	Replacement cost	Repair cost	Repair cost
Bridge approach	Replacement cost	Replacement cost	Replacement cost	Repair cost	Repair cost
Box culvert	Replacement cost	Replacement cost	Replacement cost	Repair cost	Repair cost
Pipe culvert	Replacement cost	Replacement cost	Replacement cost	Repair cost	Repair cost
Pavement	Replacement cost	Replacement cost	Replacement cost	Repair cost	Repair cost

6. Based on the scenario defined for each asset and hazard type, vulnerability values are drawn from the *Vulnerability* worksheet, as previously explained (Figure 7).

Figure 7. Scenarios for each asset under different hazards and the resulting vulnerability

Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability

7. By combining hazard probability, consequences, and vulnerability for each asset, the risk value is calculated using the following formula:

$$Risk\ value = Consequence \times Hazard\ Probability \times Vulnerability$$

8. The **risk priority** is determined by summing the risk values across different hazards and then multiplying the result by the network criticality. To help users better interpret this value, min-max normalization is applied (Figure 8). To assist users in identifying which assets require more urgent attention, the **Return on Investment (ROI)** and its **payback period** are also calculated. ROI is defined as the ratio of the maximum risk value for each asset to its direct consequence, while the payback period is the inverse of this ratio.

Figure 8. Risk Analysis Results

Risk Value[1]	Sum of Risk Value	Risk Priority[2]	Normalized Risk Priority[5]
	ROI[3]	Payback Period[4]	

• *Summary Results Worksheet*

The **Summary Results** worksheet (Figure 9) presents the key outputs of the Risk Analysis Tool. It includes asset attributes, hazard probabilities and their potential consequences, calculated risk values, risk priorities, and the corresponding ROI and payback period for each asset (It is important to note that, due to limitations in the number of cells within a single Excel file, this worksheet includes only those assets that are exposed to at least one hazard).

• *Instruction Worksheet*

The **Instruction** worksheet provides a summary of how calculations are performed in the Risk Analysis Tool. It helps users understand the underlying assumptions and formulas used throughout the process.

Worked Examples

This section presents a worked example to illustrate the use of the Risk Analysis Tool. The example utilizes data from the Utah Department of Transportation (UDOT) and is intended to help users understand how the tool can be applied within their own agencies. UDOT's goal for asset management is to foster collaboration across divisions to develop a unified program that maximizes system performance and funding efficiency, while also implementing a process to quantify risks to assets within the UDOT system. Figure 10 displays the results of the risk analysis conducted for six selected UDOT assets.

Target ID,FULLNAME	Target ID	FULLNAME	AssetType	RepairCost	ReplacementCost	AADT	TADT	Adaptive Capacity	DelayTime	Criticality_RSL	RF_Prob	RF_Direct Consequence	RF_IndirectConsequence	RF_Consequence	RF_Exposure	RF_Sensitivity
122269,HWY 6	122269	HWY 6	Approach	258808	1294040	12000	3252	High	5	0.00003172	0	258808	427033.2	685841.2	Not Close	Low
122282,HWY 6	122282	HWY 6	Approach	92285.248	461426.24	7700	3211	High	222	0.00003693	1	92285.248	152270.6592	244555.9072	Close	Low
122288,HWY 6	122288	HWY 6	Approach	560181.6	2800908	12000	3252	High	203	0.00004101	0	560181.6	924299.64	1484481.24	Not Close	Low
122289,HWY 6	122289	HWY 6	Approach	560181.6	2800908	7700	3211	High	339	0.00004242	0	560181.6	924299.64	1484481.24	Not Close	Low
122294,HWY 6	122294	HWY 6	Approach	560181.6	2800908	5800	2581	Low	1440	0.00008842	0	560181.6	924299.64	1484481.24	Not Close	Low
122298,HWY 6	122298	HWY 6	Approach	560181.6	2800908	9800	4067	High	0	0.00003365	0	560181.6	924299.64	1484481.24	Not Close	Low
122304,HWY 6	122304	HWY 6	Approach	202408.536	1012042.68	6300	2829	High	466	0.00004469	0	202408.536	333974.0844	536382.6204	Not Close	Low
122310,US 6 EB X232 OFF SR 244 RAMP	122310	X232 OFF SR 2	Approach	560181.6	2800908	0	0	High	0	0	0	560181.6	924299.64	1484481.24	Not Close	Low
122317,SOUTHERN PKWY	122317	OUTHERN PKW	Approach	376938.32	1884691.6	3000	582	High	64	0.00000958	0	376938.32	621948.228	998886.548	Not Close	Low
122321,SOUTHERN PKWY	122321	OUTHERN PKW	Approach	560181.6	2800908	2900	716	High	19	0.00000811	0	560181.6	924299.64	1484481.24	Not Close	Low
122324,STATE ST	122324	STATE ST	Approach	421696	2108480	26000	3068	High	290	0.00005948	0	421696	695798.4	1117494.4	Not Close	Low
122325,MAIN ST	122325	MAIN ST	Approach	121439.824	607199.12	2100	433	High	56	0.00000737	0	121439.824	200375.7096	321815.5336	Not Close	Low
122329,HWY 9	122329	HWY 9	Approach	560181.6	2800908	5800	1195	High	24	0.00001423	0	560181.6	924299.64	1484481.24	Not Close	Low
122330,STATE ST	122330	STATE ST	Approach	560181.6	2800908	34000	4012	High	3	0.00006014	0	560181.6	924299.64	1484481.24	Not Close	Low
122332,ZION PARK BLVD	122332	ION PARK BLV	Approach	560181.6	2800908	6600	1360	High	205	0.00002453	0	560181.6	924299.64	1484481.24	Not Close	Low
122334,ZION PARK BLVD	122334	ION PARK BLV	Approach	560181.6	2800908	6600	1360	High	204	0.00002448	0	560181.6	924299.64	1484481.24	Not Close	Low
122339,HWY 10	122339	HWY 10	Approach	560181.6	2800908	7100	3536	High	365	0.0000448	0	560181.6	924299.64	1484481.24	Not Close	Low
122340,HWY 10	122340	HWY 10	Approach	560181.6	2800908	4900	2283	High	372	0.00003567	0	560181.6	924299.64	1484481.24	Not Close	Low
122350,HWY 12	122350	HWY 12	Approach	560181.6	2800908	830	244	High	410	0.00002151	0	560181.6	924299.64	1484481.24	Not Close	Low
122352,HWY 12	122352	HWY 12	Approach	203984	1019920	2900	624	Low	1155	0.00006086	0	203984	336573.6	540557.6	Not Close	Low
122353,HWY 12	122353	HWY 12	Approach	560181.6	2800908	2900	624	Low	529	0.00003151	0	560181.6	924299.64	1484481.24	Not Close	Low
122355,MAIN ST	122355	MAIN ST	Approach	560181.6	2800908	1700	342	High	436	0.00002424	0	560181.6	924299.64	1484481.24	Not Close	Low
122358,HWY 12	122358	HWY 12	Approach	45760	228800	950	247	High	433	0.00002274	0	45760	75504	121264	Not Close	Low
122359,HWY 12	122359	HWY 12	Approach	560181.6	2800908	830	244	High	417	0.00002184	0	560181.6	924299.64	1484481.24	Not Close	Low
122362,MAIN ST	122362	MAIN ST	Approach	42239.56	211197.8	950	247	High	435	0.00002283	0	42239.56	69695.274	111934.834	Not Close	Low
122366,HWY 12	122366	HWY 12	Approach	560181.6	2800908	1600	360	High	434	0.00002414	0	560181.6	924299.64	1484481.24	Not Close	Low
122369,5200 W	122369	5200 W	Approach	590409.6	2952048	7700	3496	High	0	0.00002812	0	590409.6	974175.84	1564585.44	Not Close	Low
122371,HWY 13	122371	HWY 13	Approach	206096	1030480	8900	2376	Low	1019	0.00007094	0	206096	340058.4	546154.4	Not Close	Low
122374,4700 W	122374	4700 W	Approach	207504	1037520	2000	658	High	22	0.00000693	0	207504	342381.6	549885.6	Not Close	Low
122375,HWY 13	122375	HWY 13	Approach	560181.6	2800908	8500	1862	High	36	0.00002151	0	560181.6	924299.64	1484481.24	Not Close	Low
122376,5200 W	122376	5200 W	Approach	560181.6	2800908	1200	372	High	140	0.00000998	0	560181.6	924299.64	1484481.24	Not Close	Low
122378,HWY 14	122378	HWY 14	Approach	560181.6	2800908	1800	369	High	97	0.0000086	0	560181.6	924299.64	1484481.24	Not Close	Low
122380,HWY 14	122380	HWY 14	Approach	560181.6	2800908	1800	369	High	97	0.0000086	0	560181.6	924299.64	1484481.24	Not Close	Low
122381,CENTER ST	122381	CENTER ST	Approach	325010.4	1625052	4300	267	High	8	0.00000661	0	325010.4	536267.16	861277.56	Not Close	Low
122383,I-15 SB COL RAMP	122383	5 SB COL RAM	Approach	268576	1342880	0	0	High	0	0	0	268576	443150.4	711726.4	Not Close	Low
122391,I-15 SB FWY	122391	I-15 SB FWY	Approach	203719.472	1018597.36	26000	7800	High	2	0.00007256	0	203719.472	336137.1288	539856.6008	Not Close	Low
122403,I-15 SB FWY	122403	I-15 SB FWY	Approach	535392	2676960	45000	9945	High	8	0.00010582	0	535392	883396.8	1418788.8	Not Close	Low
122404,I-15 SB FWY	122404	I-15 SB FWY	Approach	364144	1820720	156000	22932	High	1	0.00030074	0	364144	600837.6	964981.6	Not Close	Low
122405,I-15 SB FWY	122405	I-15 SB FWY	Approach	494612.8	2473064	20000	9020	High	14	0.00007337	0	494612.8	816111.12	1310723.92	Not Close	Low
122407,I-15 SB FWY	122407	I-15 SB FWY	Approach	1560715.2	7803576	233000	25164	High	1	0.00039811	0	1560715.2	2575180.08	4135895.28	Not Close	Low
122409,I-15 SB FWY	122409	I-15 SB FWY	Approach	884400.352	4422001.76	130000	24050	High	2	0.00027843	0	884400.352	1459260.581	2343660.933	Not Close	Low
122414,I-15 SB FWY	122414	I-15 SB FWY	Approach	327360	1636800	185000	27380	High	0	0.00035763	0	327360	540144	867504	Not Close	Low
122417,I-15 SB FWY	122417	I-15 SB FWY	Approach	560181.6	2800908	25000	8775	Low	1015	0.00012444	0	560181.6	924299.64	1484481.24	Not Close	Low
122419,I-15 SB FWY	122419	I-15 SB FWY	Approach	356294.576	1781472.88	58000	12412	High	12	0.00013419	0	356294.576	587886.0504	944180.6264	Not Close	Low
122422,I-15 SB FWY	122422	I-15 SB FWY	Approach	695200	3476000	124000	22816	High	196	0.00027398	0	695200	1147080	1842280	Not Close	Low
122424,I-15 SB FWY	122424	I-15 SB FWY	Approach	241172.8	1205864	158000	29388	High	3	0.00033931	0	241172.8	397935.12	639107.92	Not Close	Low
122425,I-15 SB FWY	122425	I-15 SB FWY	Approach	454520	2272600	20000	9020	High	8	0.00007308	0	454520	749958	1204478	Not Close	Low
122431,I-15 SB FWY	122431	I-15 SB FWY	Approach	1331536.624	6657683.12	237000	30336	High	5	0.00043176	0	1331536.624	2197035.43	3528572.054	Not Close	Low
122433,I-15 SB FWY	122433	I-15 SB FWY	Approach	309759.296	1548796.48	19000	4199	High	1	0.00004457	0	309759.296	511102.8384	820862.1344	Not Close	Low
122434,I-15 SB FWY	122434	I-15 SB FWY	Approach	2618528	13092640	244000	17568	High	6	0.00036779	0	2618528	4320571.2	6939099.2	Not Close	Low
122435,I-15 SB FWY	122435	I-15 SB FWY	Approach	546639.28	2733196.4	58000	12644	High	4	0.00013512	0	546639.28	901954.812	1448594.092	Not Close	Low
122440,I-15 SB FWY	122440	I-15 SB FWY	Approach	285955.56	1429777.8	91000	22568	High	4	0.00022723	0	285955.56	471826.674	757782.234	Not Close	Low
122441,I-15 SB FWY	122441	I-15 SB FWY	Approach	222112	1110560	26000	5798	High	21	0.0000622	0	222112	366484.8	588596.8	Not Close	Low
122443,I-15 SB FWY	122443	I-15 SB FWY	Approach	777339.552	3886697.76	130000	24050	High	2	0.00027843	0	777339.552	1282610.261	2059949.813	Not Close	Low
122444,I-15 SB FWY	122444	I-15 SB FWY	Approach	990739.552	4953697.76	190000	28690	High	6	0.00037078	0	990739.552	1634720.261	2625459.813	Not Close	Low
122445,I-15 SB FWY	122445	I-15 SB FWY	Approach	560181.6	2800908	115000	28980	High	9	0.00028993	0	560181.6	924299.64	1484481.24	Not Close	Low
122446,I-15 SB FWY	122446	I-15 SB FWY	Approach	329120	1645600	56000	12656	High	4	0.00013298	0	329120	543048	872168	Not Close	Low

Figure 9. Partial View of Summary Results Worksheet

U.S. State	Utah	Unit Value of Time per passenger car ($/min)	0.3534
		Unit Value of Time per passenger truck ($/min)	1.0673

Input Data | Calculated Variables | Output Data

Asset Category	Structure ID & Full Name	Geographical Location		Asset Attributes				Risk Analysis											Results			
								Probability		Consequence				Vulnerability								
Bridge Approach	122269,HWY 6	Longitude_X	-111.589202	AADT	12000	Replacement Cost	1294040	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	427033.2	258808	427033.2	685841.2	Not Close	Low		Not Close,Low,High	0.03734016	0	0.1732218703	0.000005494597724	
				TADT	3252	Repair Cost	258808	Flood	0	3.5	427033.2	1294040	561987.243	1856027.243	Not Close	Moderate		Not Close,Moderate,High	0.01271424	0			
								Debris Flow	0	3.5	427033.2	1294040	561987.243	1856027.243	Not Close	Moderate	High	Not Close,Moderate,High	0.01271424	0	ROI[3]	Payback Period[4]	0.002869001924
		Latitude_Y	40.082508	Delay Time	5	Network Criticality *	0.00003172	Earthquake	0.01254	30	427033.2	1294040	1583782.14	2877822.14	Close	High		Close,High,High	0.0000048	0.1732218703	0.0278877698	35.85801257	
								Avalanche	0	0	427033.2	258808	427033.2	685841.2	Not Close	Low		Not Close,Low,High	0.03734016	0			
Bridge	125417,UNIVERSITY AVE	Longitude_X	-111.658754	AADT	35000	Replacement Cost	12498304	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	4124440.32	6008800	4124440.32	10133240.32	Not Close	Low		Not Close,Low,High	0.03734016	0	0.5791110867	0.000073425494 68	
				TADT	15680	Repair Cost	6008800	Flood	0	3.5	4124440.32	12498304	4430035.092	16928339.09	Not Close	Moderate		Not Close,Moderate,High	0.01271424	0			
								Debris Flow	0	3.5	4124440.32	12498304	4430035.092	16928339.09	Not Close	Moderate	High	Not Close,Moderate,High	0.01271424	0	ROI[3]	Payback Period[4]	0.03833909161
		Latitude_Y	40.208522	Delay Time	3	Network Criticality *	0.00012679	Earthquake	0.00627	30	4124440.32	12498304	6743824.08	19242128.08	Close	High		Close,High,High	0.0000048	0.5791110867	0.009653161186	103.5930076	
								Avalanche	0	0	4124440.32	6008800	4124440.32	10133240.32	Not Close	Low		Not Close,Low,High	0.03734016	0			
Box Culvert	150483,I-15 NB X222 OF…	Longitude_X	-111.840696	AADT	0	Replacement Cost	1342367	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	442981.11	268473	442981.11	711454.11	Not Close	Low		Not Close,Low,High	0.03734016	0	25.46567698	0.00001324215203	
				TADT	0	Repair Cost	268473	Flood	0	3.5	442981.11	1342367	442981.11	1785348.11	Not Close	Moderate		Not Close,Moderate,High	0.01271424	0			
								Debris Flow	0.060645	3.5	442981.11	1342367	442981.11	1785348.11	Close	Moderate	High	Close,Moderate,High	0.0002352	25.46567698	ROI[3]	Payback Period[4]	0.006914384192
		Latitude_Y	39.679984	Delay Time	11	Network Criticality *	0.00000052	Earthquake	0	30	442981.11	1342367	442981.11	1785348.11	Not Close	High		Not Close,High,High	0.00026048	0	0.08065785	12.39804929	
								Avalanche	0	0	442981.11	268473	442981.11	711454.11	Not Close	Low		Not Close,Low,High	0.03734016	0			
Pipe Culvert	125691,HWY 96	Longitude_X	-113.140813	AADT	5000	Replacement Cost	655030	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	216159.9	131006	216159.9	347165.9	Not Close	Low		Not Close,Low,High	0.03734016	0	5.658700556	0.00008374876823	
				TADT	1145	Repair Cost	131006	Flood	0	3.5	216159.9	655030	854323.8898	1509353.89	Not Close	Moderate		Not Close,Moderate,High	0.01271424	0			
								Debris Flow	0.01594	3.5	216159.9	655030	854323.8898	1509353.89	Close	Moderate	High	Close,Moderate,High	0.0002352	5.658700556	ROI[3]	Payback Period[4]	0.043729384 61
		Latitude_Y	37.10722	Delay Time	61	Network Criticality *	0.0000148	Earthquake	0	30	216159.9	655030	5686136.955	6341166.955	Not Close	High		Not Close,High,High	0.00026048	0	0.03672976963	27.22587182	
								Avalanche	0	0	216159.9	131006	216159.9	347165.9	Not Close	Low		Not Close,Low,High	0.03734016	0			
Other Culverts	151038,HWY 35	Longitude_X	-110.432789	AADT	530	Replacement Cost	780609	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	257600.97	156122	257600.97	413722.97	Not Close	Low		Not Close,Low,High	0.03734016	0	0.1329112087	.0000005582270766	
				TADT	126	Repair Cost	156122	Flood	0	3.5	257600.97	780609	327427.6206	1108036.621	Not Close	Moderate		Not Close,Moderate,High	0.01271424	0			
								Debris Flow	0.00051	3.5	257600.97	780609	327427.6206	1108036.621	Close	Moderate	High	Close,Moderate,High	0.0002352	0.1329112087	ROI[3]	Payback Period[4]	0.0002914780366
		Latitude_Y	40.270859	Delay Time	62	Network Criticality *	0.0000042	Earthquake	0	30	257600.97	780609	856115.118	1636724.118	Not Close	High		Not Close,High,High	0.00026048	0	0.000723920268	1381.367596	
								Avalanche	0	0	257600.97	156122	257600.97	413722.97	Not Close	Low		Not Close,Low,High	0.03734016	0			
Pavement	6,HWY 12	Longitude_X	-112.178084	AADT	2900	Replacement Cost	1315987.2	Hazard Type	Hazard Probability	Required Repair Days	Impeded Flow Cost	Direct Consequence	Indirect Consequence	Consequence	Exposure	Sensitivity	Adaptive Capacity	Scenario Description	Vulnerability	Risk Value[2]	Sum of Risk Value	Risk Priority[4]	Normalized Risk Priority[5]
								Rockfall	0	0	434275.776	263197.44	434275.776	697473.216	Not Close	Low		Not Close,Low,Low	0.03814272	0	10.3318497	0.000325555658 41	
				TADT	624	Repair Cost	263197.44	Flood	0.01	3.5	434275.776	1315987.2	3564894.179	4880881.379	Close	Moderate		Close,Moderate,Low	0.00021168	10.3318497			
								Debris Flow	0	3.5	434275.776	1315987.2	3564894.179	4880881.379	Not Close	Moderate	Low	Not Close,Moderate,Low	0.012936	0	ROI[3]	Payback Period[4]	0.1699892354
		Latitude_Y	37.696931	Delay Time	529	Network Criticality *	0.00003151	Earthquake	0	30	434275.776	1315987.2	27268147.8	28584135	Not Close	High		Not Close,High,Low	0.00025872	0	0.03708912502	26.96208119	
								Avalanche	0	0	434275.776	263197.44	434275.776	697473.216	Not Close	Low		Not Close,Low,Low	0.03814272	0			

*Network criticality is obtained from the criticality tool.
[2]Risk value = Consequence × Hazard Probability × Vulnerability
[3]Risk priority = Risk value × Network Criticality
[3]ROI=Risk Value/Direct Consequence
[4] Payback Period=1/ROI
[5] Min-Max Normalized Risk Priority=(Risk Priority - Minimum Risk Priority in Data Set) / (Maximum Risk Priority in Data Set - Minimum Risk Priority in Data Set)*100

Figure 10. Risk Analysis Tool for Selected Assets of UDOT